



University of Pisa

Department Of Computer Science

Data Mining

Data Mining Project Report

Rap Italian Songs

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1 Data Understanding and Preparation

For this project, we worked on a dataset of Italian rap songs, which includes information about tracks, lyrics, audio features, artist metadata, etc. The dataset is divided into two CSV files: one containing over 11,000 tracks and 44 features, and another describing over 100 artists and 14 features.

In this chapter, we outline the end-to-end preprocessing pipeline designed to transform the raw datasets into a structured format suitable for analysis. The workflow follows a sequential logic, starting with the refinement of the **Artists** dataset, where we employ a retrieval-augmented strategy (using APIs and LLM agents) to reconstruct missing biological and geographical contexts.

Subsequently, the focus shifts to the **Tracks** dataset, which undergoes a rigorous Single Feature Analysis. This standardized protocol applies five strict phases—data quality assessment, missing value imputation, distribution transformation, outlier detection, and post-processing (winsorization and scaling)—to every attribute. Crucially, this pipeline is dynamic: **Feature Engineering** is integrated mid-process, deriving new indicators (like `swear_ratio`) immediately after the raw data imputation and before the final statistical validation. The process concludes with a holistic Multivariate Analysis, addressing feature correlations and complex irregularities across the entire data space.

1.1 Artists Dataset: Analysis and Preparation

The refinement of the `artists` dataset employed a multi-layered pipeline to reconstruct biological data and enforce relational consistency. To address missing attributes, we utilized the `ArtistImputer` module with a hierarchical logic:

1. **Wikidata Knowledge Graph:** For structured attributes like `gender`, `birth_date`, and `active_start`.
2. **LLM-Assisted Extraction:** For unstructured data (e.g., `region`), using a Gemini agent to parse Wikipedia biographies.

To ensure data integrity, we validated the process by comparing feature distributions (e.g., birth years) **before and after imputation**. This visual inspection confirmed that gaps were filled effectively without introducing statistical bias or distorting the underlying population structure.

Following retrieval, we validated attributes against logical constraints (e.g., $\text{Age} \in [10, 100]$). Crucially, completeness for `region` and `birth_year` was enforced: artists still missing these fields were discarded. To preserve referential integrity, we performed a cascading deletion, removing all tracks linked to these incomplete profiles.

1.2 Feature Engineering

Before initiating the statistical cleaning pipeline, we expanded the initial feature space by deriving meaningful numerical indicators from the raw attributes. By combining low-level signals into high-level semantics, we aimed to capture complex patterns suitable for the subsequent analysis.

For the sake of narrative clarity, we define the full feature set here. However, to ensure data robustness, these engineered indicators were computationally derived only after the imputation of missing values in the raw attributes (e.g., lyrics, dates).

Mood & Content Metrics

- **swear_ratio:** The density of explicit content, defined as the fraction of profanities (IT/EN) relative to the total token count:

$$\text{swear_ratio} = \frac{\text{swear_EN} + \text{swear_IT}}{\text{n_tokens}}$$

- **aggressiveness:** A composite index representing track intensity. It is calculated as the mean of normalized high-energy spectral features (**centroid**, **rolloff**, **loudness**, **flux**, **bpm**) combined with **swear_ratio**.

Temporal & Biographical Metrics

We parsed metadata to place songs in their historical context:

- **song_age:** Years elapsed since the track's release:

$$\text{song_age} = \text{current_year} - \text{release_year}$$

- **artist_age_at_release:** The artist's biological age at the time of the song's debut, useful for analyzing demographic trends:

$$\text{artist_age_at_release} = \text{release_year} - \text{birth_year}$$

- **release_season:** A categorical mapping of release months to seasons, utilized to detect seasonal industry strategies (e.g., "summer hits").

Contextual Metrics

To measure success relative to the artist's specific career path:

- **relative_popularity:** A rank-based metric that contextualizes a song's success within the artist's own repertoire, isolating their top-performing tracks regardless of global fame.
- **has_collaboration:** A binary flag indicating guest appearances. This serves as a proxy for the "network effect," where collaborative tracks leverage multiple fanbases to boost performance.

1.3 Data Quality Assessment

We implemented a strict auditing protocol to identify and mask erroneous data prior to imputation. This assessment enforced domain-specific constraints across four key dimensions:

- **Structural Integrity:** We addressed sparsity patterns and enforced strict uniqueness by verifying composite keys (e.g., Artist + Title) to eliminate subtle redundancies.
- **Value Validity:** We applied physical boundaries to all features. Categorical variables were validated against allowed sets, while numerical attributes (e.g., bpm, duration) were checked against logical ranges to filter impossible values.
- **Temporal Consistency:** We cross-referenced release dates with artist metadata. Entries were flagged if the track predated the artist’s `active_start`, implied an artist age < 15, or fell outside the genre’s scope (pre-1990).
- **Textual Quality:** To ensure NLP reliability, we filtered lyrics that were implausibly short (< 100 characters) or purely repetitive.

Outcome: Values failing these checks were systematically set to missing (NaN). This prevented noise propagation, preparing the invalid entries for reconstruction during the subsequent *Imputation* phase.

1.4 Handling Missing Values in Tracks

Following the elimination of invalid entries, our approach prioritized external data retrieval over statistical estimation. We implemented a two-phase pipeline: first querying a multi-source API ecosystem to reconstruct ground-truth data, and then applying context-aware imputation only for residual gaps.

1.4.1 Phase 1: Multi-Source API Retrieval

We integrated specialized APIs to reconstruct missing metadata and content:

- **Metadata and Metrics:** We leveraged MusicBrainz as the primary source for album information and release dates, augmented by Spotipy (Spotify API) to retrieve popularity scores and verify track durations.
- **Textual Content:** The GeniusLyrics API was utilized to scrape and verify the lyrics feature, ensuring high-quality input for downstream NLP tasks.

1.4.2 Phase 2: Statistical Fallback Strategies

For data points that remained missing after the retrieval phase, we applied a tiered imputation pipeline based on domain logic.

Context-Aware Hierarchical Imputation For continuous audio descriptors (e.g., loudness, bpm), we avoided simple global averages. Instead, we implemented a three-level fallback mechanism:

1. **Album Context:** Imputation via the track’s album mean (preserving production consistency).
2. **Artist Context:** Fallback to the artist’s mean if album data is unavailable.

3. **Global Context:** Global dataset mean used only as a final safety net.

Logic-Based & Correlation Imputation Specific features required tailored reconstruction strategies:

- **Lexical Features:** Statistics like `n_tokens` were re-computed directly from the raw lyrics column whenever available, prioritizing consistency over estimation.
- **Duration:** Missing values were imputed based on the strong correlation with lyrics length (using `n_sentences` bins).
- **Categorical Defaults:** `album_type` was defaulted to the mode ("single"), while the `explicit` flag was inferred by scanning the lyrics against a weighted keyword dictionary.

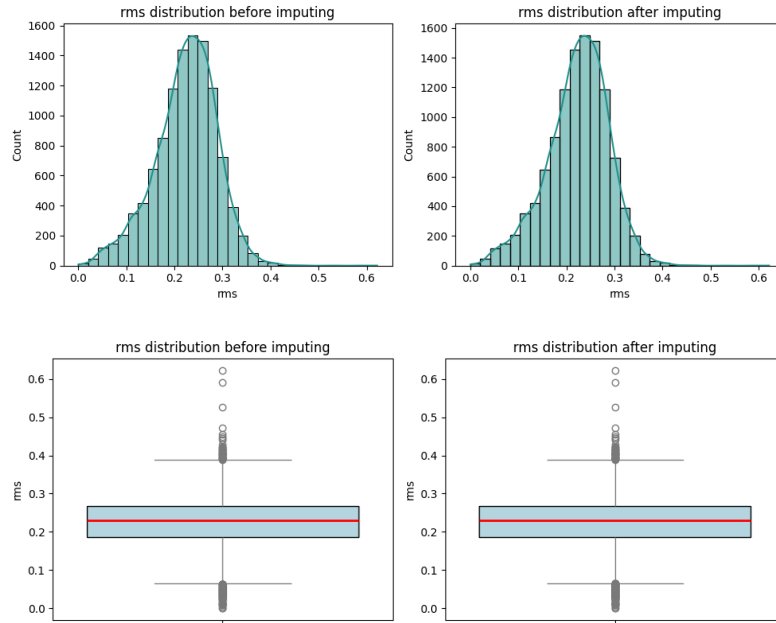


Figure 1: Validation of imputation strategy: comparison of feature distribution (top) and statistical quartiles (bottom) before and after the process.

1.4.3 Phase 3: Imputation Validation

To ensure data integrity, the entire process was monitored by systematically comparing feature distributions before and after imputation . We utilized histograms to check for shape distortion (e.g., artificial peaks) and boxplots to verify statistical consistency (Median/IQR). Strategies were accepted only if they filled gaps without altering the feature’s original statistical properties.

Imputation Cut-off We enforced a strict threshold for excessive missingness: features exceeding a critical sparsity limit were discarded rather than imputed to prevent synthetic bias. For instance, `stats_pageviews` was removed due to extreme gaps (>40% missing), as statistical reconstruction would have severely compromised the model’s integrity.

1.5 Feature Transformation

To satisfy the normality assumptions of distance-based algorithms, we systematically evaluated four transformation candidates to minimize skewness and kurtosis:

- **Identity:** Baseline (no change).
- **Logarithmic** ($\log(x + 1)$): For compressing strong right tails.
- **Square Root:** For handling mild right skewness.
- **Yeo-Johnson:** An optimized power transformation for variance stabilization.

Selection relied on minimizing absolute skewness with a strict "simplicity bias": complex methods (like Yeo-Johnson) were discarded if they offered marginal improvement ($< 5\%$) over simpler options to preserve interpretability.

As illustrated in Figure 2, the evaluation included a visual inspection of both histograms (distribution shape) and boxplots (outlier compression). In the specific example shown, the Identity form was retained as the original distribution was already sufficiently Gaussian.

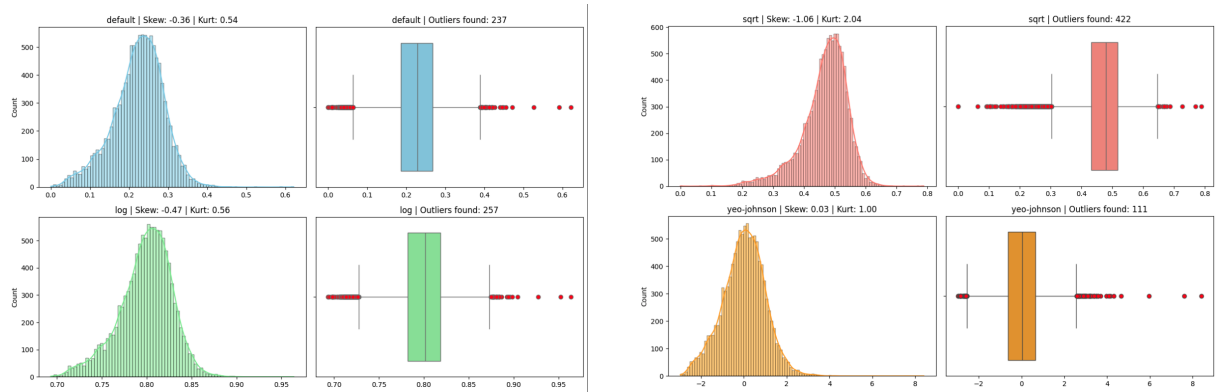


Figure 2: Comparison of transformations. The plots display histograms (top) and boxplots (bottom) for different techniques. Left: Default and Log. Right: Sqrt and Yeo-Johnson. In this case, no transformation was applied as the data was already normal.

Method	Features Selected
Log	bpm
Yeo-John.	rms, tokens_per_sent, avg_token_per_clause, year, swear_IT, swear_EN, swear_ratio
Sqrt	zcr, rolloff, duration_sec, song_age
Identity	flux, loudness, spectral_complexity, flatness, pitch, centroid, n_tokens, n_sentences, lexical_density, char_per_tok, popularity, aggressiveness, starting_age, relative_popularity, artist_age, artist_age_at_release, release_season, has_collaboration, artist_collab_rate

Table 1: Summary of optimal transformations.

1.6 Univariate Outlier Detection

Anomaly detection was applied directly to the transformed distributions (e.g., post-Yeo-Johnson) rather than raw data. This ensures that flagged values represent genuine statistical deviations persisting after skewness correction.

We employed the robust Interquartile Range (IQR) method, identifying outliers as data points falling outside standard statistical fences:

$$\text{Outlier} \iff x_{trans} < (Q_1 - 1.5 \cdot IQR) \quad \vee \quad x_{trans} > (Q_3 + 1.5 \cdot IQR)$$

1.7 Post-Processing: Winsorization & Scaling

To mitigate the influence of outliers without reducing dataset size, we applied Winsorization. Instead of deletion, extreme values were capped at the IQR fences ($Q_{1/3} \pm 1.5 \cdot IQR$), effectively suppressing their variance impact while preserving their rank order.

Subsequently, to prevent magnitude bias in distance-based algorithms, we normalized the feature space using StandardScaler. All continuous attributes were transformed into Z-scores ($z = \frac{x-\mu}{\sigma}$), ensuring that physically disparate measures (e.g., `duration_ms` vs. `speechiness`) contribute equally to the multivariate analysis.

1.8 Feature Selection

We filtered the datasets to remove identifiers, redundant fields, and low-quality features:

- **Tracks:** Dropped `id`, `id_album`, and `album_image` (non-analytical); `full_title`, `primary_artist`, and `album` (redundant); `modified_popularity` (zero variance); and `stats_pageviews` (excessive missingness).
- **Artists:** Excluded `description` (unstructured), `active_end` (100% missing), and `nationality` (near-zero variance, as the dataset is almost exclusively Italian).

1.8.1 Multicollinearity Analysis

To assess feature independence, we compared Pearson (linear) and Spearman (monotonic) matrices. The results were remarkably consistent (Figure 3), confirming the robustness of the relationships.

We identified strong multicollinearity ($|r| > 0.8$) in specific pairs and applied a selection strategy to reduce redundancy:

- `n_sentences` vs. `n_tokens` (0.87) → Dropped `n_sentences`.
- `rolloff` vs. `zcr` (0.97) → Dropped `zcr`.
- `rms` vs. `loudness` (0.99) → Dropped `rms`.

Conversely, the artists dataset required no such analysis as it contains only spatial coordinates (latitude/longitude).

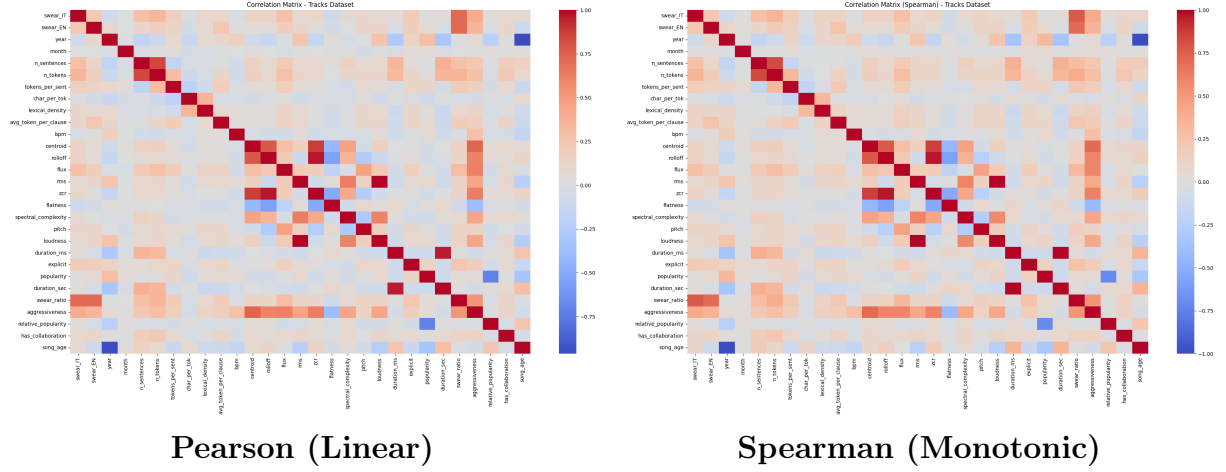


Figure 3: Side-by-side comparison of correlation matrices. The visual similarity confirms that feature relationships are robust across both linear and rank-based metrics.

1.9 Multivariate Outlier Detection

To detect inconsistencies arising from feature *combinations* (e.g., high energy yet zero loudness), we adopted a holistic approach, benchmarking four unsupervised algorithms: **Isolation Forest**, **Local Outlier Factor (LOF)**, **One-Class SVM**, and **GMM**.

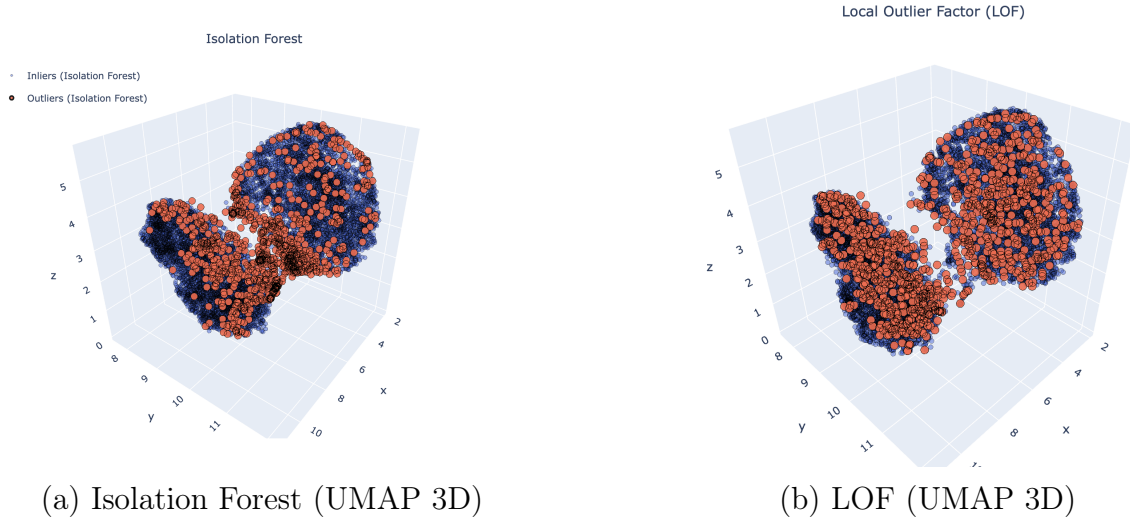


Figure 4: 3D UMAP projection of multivariate outliers. Isolation Forest (left) identifies peripheral anomalies more cleanly than LOF (right), which struggles with variable density clusters. Both models were fixed to flag exactly 3% of the data.

To ensure a fair comparison, we implemented a fixed-budget strategy. We established a baseline using Isolation Forest with a contamination of 0.03 (3%). For the other models, we computed continuous anomaly scores and dynamically calibrated the thresholds to flag the exact same number of instances. This isolated the comparison to the quality of the decision boundaries rather than sensitivity settings.

Validation and Outcome: We validated the models by projecting the high-dimensional

feature space into both 2D and 3D using a suite of complementary reduction techniques: PCA (as a linear baseline), t-SNE (for local structure), and UMAP (for global topology). This multi-view inspection (Figure 4) confirmed that Isolation Forest provided the most robust separation, correctly identifying points in sparse peripheral regions without over-flagging dense clusters. Unlike univariate outliers (which were Winsorized), these multivariate anomalies represent structural noise and were dropped entirely to ensure the downstream clustering operates on a cohesive core.

2 Clustering Analysis

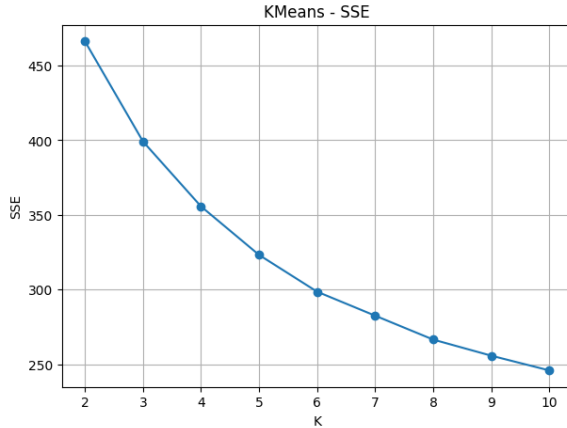
2.1 K-means

2.1.1 Pre-processing

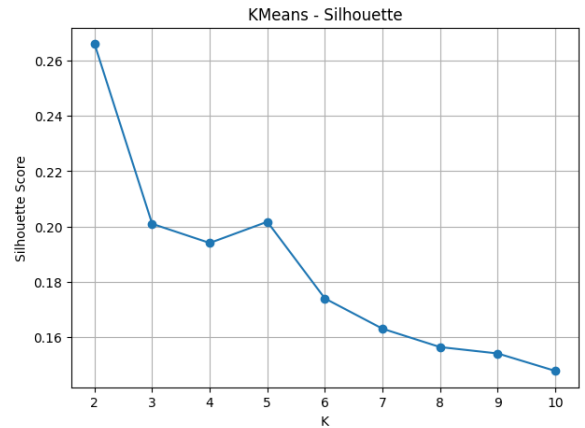
We selected a subset of informative features for clustering, including audio features (e.g., centroid, rolloff), popularity, and engineered features such as aggressiveness. Because K-Means is sensitive to differences in scale, we applied feature scaling and tested two normalization methods (Z-Score and Min-Max) and selected the most suitable one for each algorithm, for example we found that K-means performed better with Min-Max.

2.1.2 Finding best K

The best value of k was found by running the K-means algorithm 10 times, in which we were able to compute the following silhouettes and SSE plots.



(a) K-Means SSE over K clusters



(b) K-Means Silhouette score

Figure 5: K-Means clustering evaluation metrics

We choose $k = 3$ by following the informal elbow rule for the SSE graph, which had a Silhouette score of 0.20. We can notice that for $k > 3$ the Silhouette score starts to drop.

2.1.3 Understanding the Clusters

After fitting K-Means with $k = 3$, we analyzed the clusters and found:

Cluster 0 – High-Energy / Complex Tracks

Higher loudness (+4.5 than average), spectral complexity (+8.78), rolloff (+450), lyric length (+36 tokens), and aggressiveness (+0.41), with slightly below-average popularity. Represents loud, energetic tracks (e.g., trap).

Cluster 1 – Soft / Simple / Calm Tracks

Lowest loudness (−10), complexity (−10.37), rolloff (−221), lyric length (−66), and aggressiveness (−0.47), with slightly below-average popularity. Represents calm, simple tracks (e.g., acoustic).

Cluster 2 – Mainstream / Balanced Tracks

Almost all features are near the global mean, with highest popularity. Represents tracks that don't have extreme characteristics (mainstream).

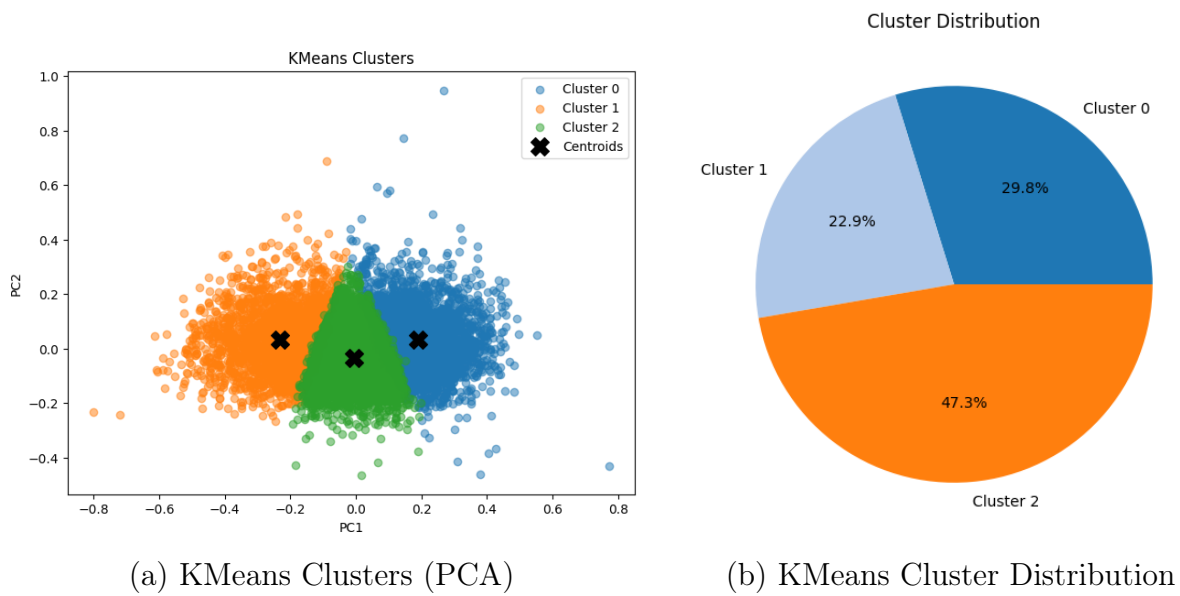


Figure 6: KMeans cluster visualization and distribution

2.2 Density Based Clustering

2.2.1 Pre-processing

Like before, we applied the same pre-processing logic to our dataset, which included: features selection and scaling using different methods to see which one performs best.

2.2.2 Finding Best Parameters

In DBSCAN, we have to choose two parameters **eps** and **min_samples**. We used a combination of the k-distance graph (knee method) and grid search to select the best parameters, this way our choice will be solid and well-thought and not just guessed or lucky.

For **min_samples** we used the heuristic $2 \times \text{dimension}$ as our starting point, and we created a search range around it. For example, if we assume $D=7$ in our case, then the search range will be around that number (e.g., 9 to 20).

For **eps**, we plotted the K-Distance Graph, and used the knee method which visually suggested that a good range for **eps** was between 0.17 and 0.28.

Then, we performed a Grid Search across the ranges we found, by running the DBSCAN algorithm using each parameter combination (60 different combinations) and evaluating the results using the Silhouette Score, which measures how well-separated and distinct the clusters are. Of course, we excluded (did not compute the Silhouette Score) for results with only 2 clusters (i.e., main cluster + noise) since they are useless to us.

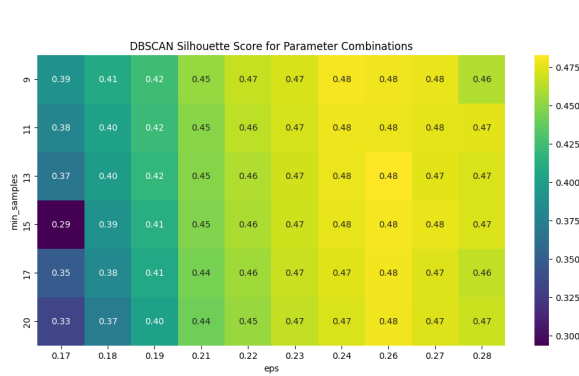


Figure 7: DBSCAN Silhouette Scores

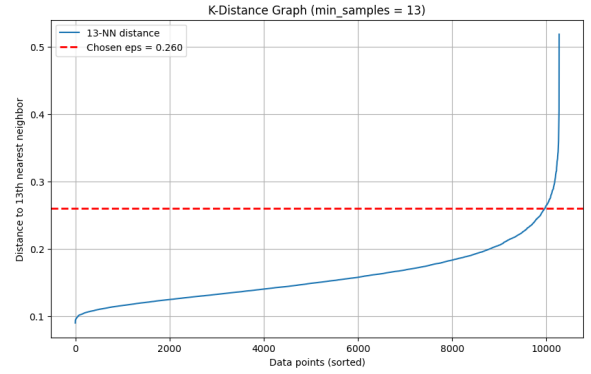


Figure 8: K-Distance Graph of the Best Parameters

After this thorough search process, the best parameter combination found that maximized the Silhouette Score at 0.4831 is: $\epsilon = 0.260$ and $MinPts = 13$.

To validate these results, we re-used the knee method one last time to make sure our results are coherent and visually correct, and they were.

We tried the search grid on both normalized data sets, using MinMax first, which gave us a Silhouette Score up to 0.4831, then using Z-Score scaling, which gave us a Silhouette Score up to 0.1821, and after reviewing both results we decided that MinMax scaling is the better option in this case.

2.2.3 Characterization and Interpretation of the Clusters

The best DBSCAN model grouped the tracks into two main clusters and flagged a small number of noise points (21 tracks only).

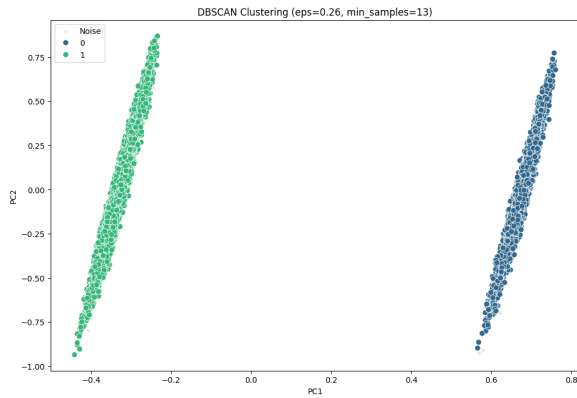


Figure 9: DBSCAN PCA

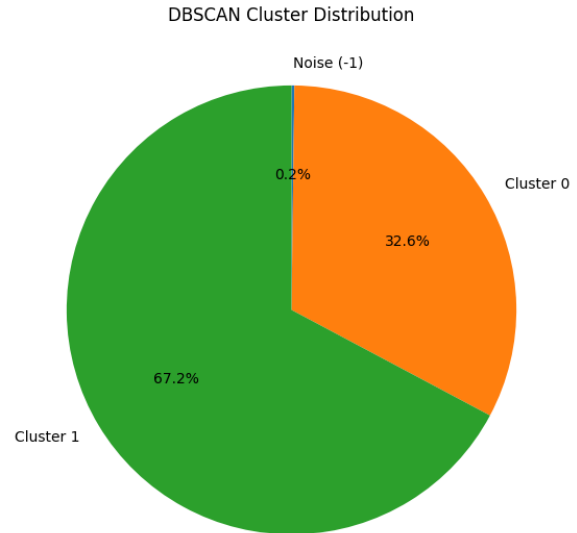


Figure 10: DBSCAN Distribution

Cluster 0 – Collaborative and High-Energy Tracks (32.6%)

The `has_collaboration` is exactly 1.00 (all tracks are collaborations), highest loudness (25.85), spectral complexity (28.98), and aggressiveness (0.123).

Cluster 1 – Solo and Smooth Tracks (67.2%)

The `has_collaboration` is 0.00 (all tracks are solo), lowest loudness and near-zero aggressiveness, indicating less intense tracks.

Cluster -1: Noise (0.20%)

Lowest spectral complexity (21.9) and aggressiveness (-0.07) but highest popularity (54.52). These are unique tracks that achieved viral success.

2.3 Hierarchical Clustering

2.3.1 Pre-processing

Hierarchical clustering was performed using the Agglomerative Clustering approach. The used linkage methods are: Ward, Single, Complete, Average.

To ensure fair comparison, we used the same features as in previous algorithms. Min-Max and Z-Score normalization were tested, but Z-Score gave slightly better cluster balance and interpretability.

2.3.2 Ward Method

The Ward linkage method produced well-balanced clustering results. The dendrogram shows a clear separation at higher linkage distances, which means the data was naturally split into two main groups.

To confirm this observation, we tested different values of k , and found that $k=2$ gave the best silhouette score (0.1970). The cluster size distribution is fairly balanced.

Analyzing the cluster profiles, one cluster had higher loudness, higher spectral complexity, and higher aggressiveness compared to the global mean, suggesting more intense and energetic tracks. The other cluster shows lower values for these features, indicating calmer and less aggressive songs.

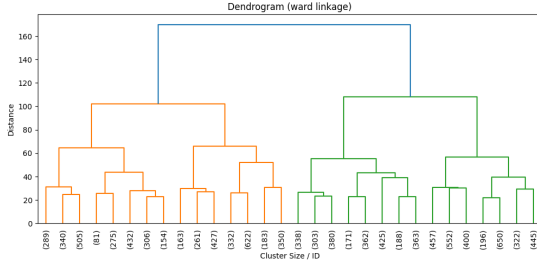


Figure 11: Dendrogram Ward

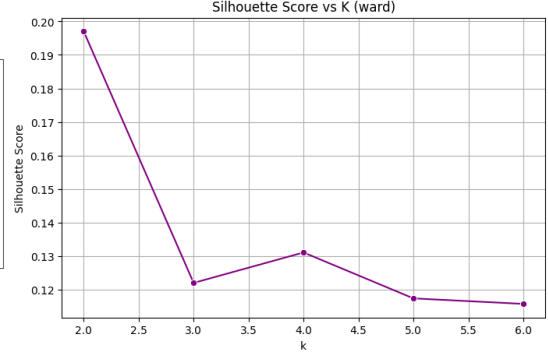


Figure 12: Silhouette Score Ward

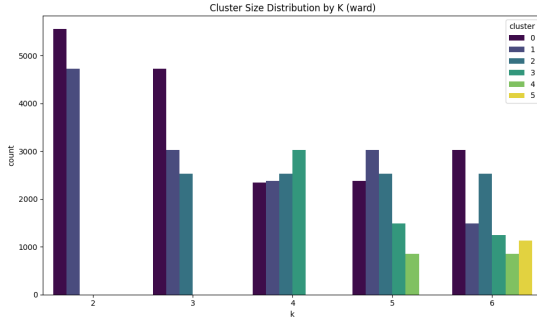


Figure 13: Cluster Size Distribution Ward

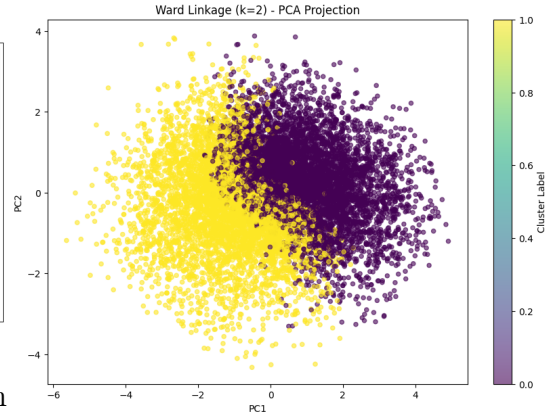


Figure 14: PCA Ward

2.3.3 Single Method

Although single linkage achieved a relatively high silhouette score (0.3925), the dendrogram and cluster distribution reveal a problematic clustering structure, with most data points merged into one large cluster and a single-song cluster. Cluster profiles show that the small cluster is an extreme outlier (very high loudness and spectral complexity and extremely low popularity), while the large cluster closely matches the global mean. Therefore, single linkage fails to capture meaningful structure and is unsuitable for this analysis.

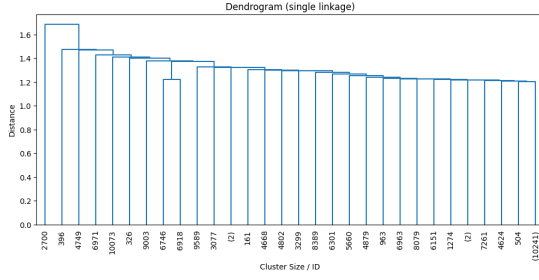


Figure 15: Dendrogram Single

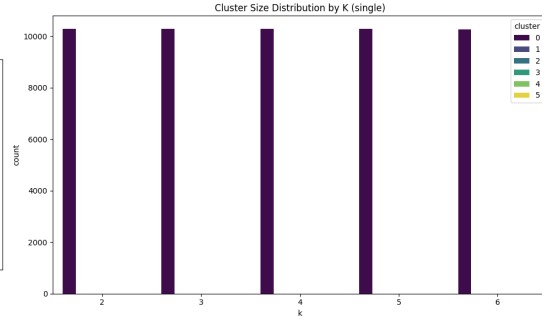


Figure 16: Cluster Size Distribution Single

2.3.4 Complete Method

Complete linkage gave us more balanced and interpretable results compared to single linkage. The dendrogram shows clearer separations than single linkage, but still less than Ward linkage. The silhouette analysis indicates that $k=2$ is again the best clustering, with a silhouette score of 0.1951. From the cluster profiles, one cluster is more aggressive, louder, and spectrally complex than the global mean, while the other cluster shows lower values for these features.

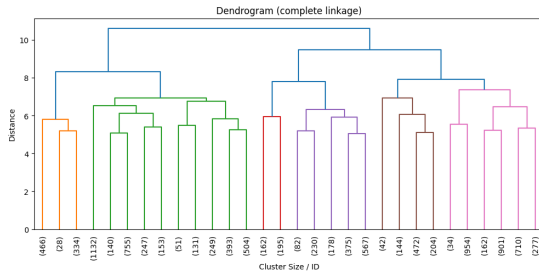


Figure 17: Dendrogram Complete

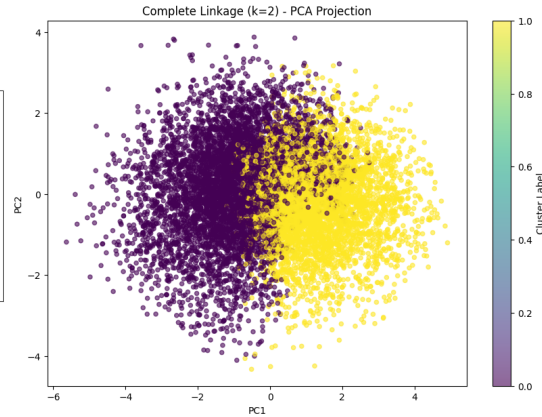


Figure 18: PCA Complete

2.3.5 Average Method

Average linkage shows similar limitations to single linkage, although the silhouette score for $k=2$ is relatively high (0.3925), the dendrogram and size distribution show a highly unbalanced structure. One cluster contains nearly all the data points, while the other cluster is composed of only one song. So this method is also not suitable for this analysis.

2.4 X-Means

A major challenge of the K-Means algorithm is pre-defining the number of clusters (k), which often relies on subjective visual interpretations of "elbow" plots. To address this issue, we applied X-Means clustering from the pyclustering package, which is extended version of K-Means that automatically estimates the optimal number of clusters.

The algorithm starts with a minimum k (in our case it was 2) and determines whether to split a cluster into two sub-clusters by evaluating the Bayesian Information Criterion (BIC).

2.4.1 Results

X-means automatically identified the optimal number of clusters as 2, which gave us a Silhouette Score, of 0.2356.

To interpret the resulting clusters, we analyzed the deviation of each cluster’s mean feature values from the global mean. The first cluster had higher-than-average popularity, and lower loudness, spectral complexity, and aggressiveness. While the other cluster had higher loudness, spectral complexity, rolloff, and aggressiveness, but lower popularity than average. So we can say that the first cluster has more popular and smoother tracks, while the second has louder, more aggressive, but less popular tracks.

2.5 Comparison of the Clustering Algorithms

Algorithm	Clusters	Silhouette	Cluster Profiles
K-Means	3	0.20	High-Energy Complex Tracks vs Soft Calm Tracks vs Mainstream Balanced Tracks
DBSCAN	2 (+ noise)	0.4831	Collaborative and High-Energy Tracks vs Solo and Smooth Tracks (+ Noise)
Hierarchical (Ward)	2	0.197	High-Intensity Aggressive Tracks vs Low-Intensity Calm Tracks
X-Means	2 (auto)	0.2356	Popular and Smooth Tracks vs Loud and Aggressive Tracks

Table 2: Comparison of the clustering algorithms

Overall, the clustering results obtained using the different algorithms are largely consistent. Most methods identify a separation between louder and more aggressive VS calmer and smoother tracks. If we evaluate the algorithms based on their Silhouette score, DBSCAN had the best cluster separation since it achieved the highest score compared to the other methods.

3 Predictive Analysis & XAI

3.1 Dataset Statistics

We created the following five classes:

1. **Campania:** Campania, Molise
2. **Lazio:** Lazio, Toscana, Marche, Umbria, Abruzzo, Emilia-Romagna

3. **Lombardia:** Lombardia

4. **North:** Piemonte, Veneto, Valle d’Aosta, Trentino, Liguria, Friuli-Venezia-Giulia

5. **South:** Sicilia, Sardegna, Puglia, Basilicata, Calabria

Based on domain knowledge, we know that Roma, Napoli and Milano are the cities that form the main school of origin for the Italian rap genre; for this reason we decided to create the ad-hoc classes Lombardia, Lazio and Campania.

Initially we also had a class Center, but we decided to merge it in the class Lazio, because it created a big imbalance in the class distribution: Center contained only 113 instances, while Lombardia had over 3000.

Not all artists were suitable for the task, because their region of origin was missing data that we couldn’t reliably infer. Thus, the dataset is composed of 8706 songs, from 77 artists.

3.1.1 Class Distribution

The class ratios are contained in Table 3

Class	Campania	Lazio	Lombardia	North	South
Tracks	933	1910	3446	1282	1135

Table 3: Class Distributions

3.1.2 Noise and Task Complexity

The dataset likely contains some noise, partly due to the imputation performed during the data preparation phase. Moreover, the region of origin of the artist does not necessarily imply an adherence to that specific school of rap; for instance, a rapper born in Napoli might be influenced by the style of the Milano scene and choose to imitate it.

For this reason we think that the class labels are not well-separated in the feature space and we cannot expect excellent classification results.

3.2 Features

We created only one new feature, the rest of the features are the same used for the clustering: they are all numerical and mostly focus on the lyrical and musical properties, which should reasonably characterize the school of origin of the rapper.

3.2.1 Regional References

Rappers often reference their hometowns or neighborhoods, sometimes dedicating entire songs to these places. Furthermore, some rap schools encourage the use of local dialects, with Napoli and Rome being prominent examples. For these reasons, we created a categorical feature, **region_references**, indicating the class most referenced in the lyrics. The possible references were hard-coded in `task3/utils.py` using domain knowledge and online sources on Italian dialects. It is a quite sparse feature, since only $\sim 19.6\%$ of the

tracks have a value for it, but it still turned out to be useful for classification (section 3.6). Table 4 shows the feature distribution.

No-reference	Campania	Lazio	Lombardia	North
6997 ($\sim 80.4\%$)	503 ($\sim 5.8\%$)	491 ($\sim 5.6\%$)	666 ($\sim 7.6\%$)	53 ($\sim 0.6\%$)

Table 4: Region Reference Distribution

3.3 Data Preprocessing and Dealing with Imbalance

3.3.1 Preprocessing

- Standard scaling for numerical features
- One hot encoding for categorical features: `region_references` and `release_season`.

3.3.2 Class Imbalance

The class Lombardia contains approximately 3.5 times the number of instances of the Campania class.

While it is not a case of pronounced imbalance, we tested several methods for dealing with it nonetheless.

In particular, we experimented with adjusting the parameter `class_weight` of the models, as well as different sampling strategies, namely `RandomOverSampler`, `RandomUnderSampler` and `SMOTE`.

All of them were included as part of the model selection procedure, section 3.4.1.

None of the sampling methods turned out to be effective. For the reasons discussed in section 3.1.2, we can expect that oversampling introduced more noise, while undersampling cut out valuable instances that could have been useful during learning.

3.4 Models Tested and Selection Procedure

We divided the dataset in test set ($\sim 20\%$ of the data; used only for assessing the capabilities of the final model) and train-validation set ($\sim 80\%$ of the data). Both sets maintained the class distribution ratios

We tested the following models:

1. **Linear Model:** `RidgeClassifier` (sklearn).
2. **Decision Tree:** `DecisionTreeClassifier` (sklearn).
3. **K-Nearest Neighbors:** `KNeighborsClassifier` (sklearn)
4. **Support Vector Machine:** `SVC` (sklearn)
5. **Random Forest:** `RandomForestClassifier` (sklearn)
6. **Gradient Boosting:** `lightGBMClassifier` (lighthGBM)
7. **Explainable Boosting Machine:** `ExplainableBoostingClassifier` (interpret)

The first three are simple baseline models, while the last four are complex models.

3.4.1 Model Selection

We used the train-validation set with `RandomizedSearchCV` from `Scikit-Learn` to efficiently search hyperparameters, combined with stratified 5-fold cross-validation to preserve class ratios. The best-performing model was then further tuned using `GridSearchCV`. Finally, the resulting model with optimized hyperparameters was retrained on the entire train-validation set and evaluated on the test set.

3.4.2 Metrics Used

The main metrics we used are `balanced_accuracy` and `f1_weighted`, because of the class imbalance.

We also included `accuracy` for completeness, but we didn't rely on it.

3.5 Results

3.5.1 Baseline Models Results

The results of the cross validation for the baseline models are illustrated in table 5.

Model	Balanced Accuracy	F1 Weighted	Accuracy
Decision Tree	0.451892	0.405478	0.405132
KNN	0.465222	0.429093	0.423796
Linear Model	0.436093	0.38547	0.384471

Table 5: Baseline Models Mean Cross Validation Results

3.5.2 Complex Models Results

The results of the validation of the complex models are illustrate in table 6.

Model	Balanced Accuracy	F1 Weighted	Accuracy
SVM	0.509627	0.481013	0.475459
Gradient Boosting	0.612171	0.654011	0.658298
Random Forest	0.534278	0.514898	0.508177
Explainable Gradient Boosting	0.606252	0.647276	0.653127

Table 6: Complex Models Randomized Search with Cross Validation Results

3.5.3 Final Model

We chose Explainable Gradient Boosting as the final model because of its explainability capacity and because its score turned out very close to Gradient Boosting. Finally, we performed a final grid search on it to find the best hyperparameter combination. We then tested it on the test set. The results are shown in table 7. We also computed the confusion matrix, illustrated in figure 20.

	Balanced Accuracy	F1 Weighted	Accuracy
Validation	0.634749	0.669817	0.673364
Test	0.625075	0.659915	0.661883

Table 7: Final Model Results

As expected, because of the class imbalance, the most common misclassification error consists in labelling as Lombardia a different class.

3.6 Explainability

Thanks to the inherent explainability of EBM we gain some valuable insights on its inner workings. An overview of the feature importances is provided by the summary plot in Figure 19. The model seems to heavily rely on **popularity** and **relative_popularity**.

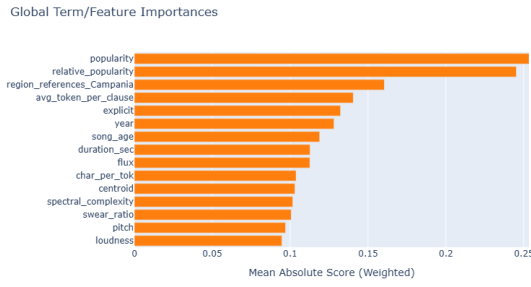


Figure 19: Summary Plot of EBM

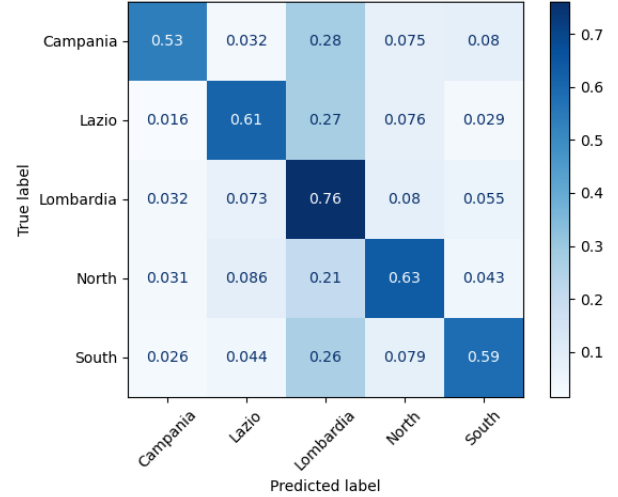


Figure 20: Confusion Matrix

Some musical features, like **avg_token_per_clause**, **spectral_complexity** and **flux**, are also present, all with a similar importance.

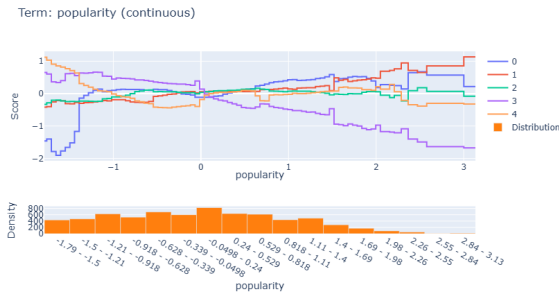


Figure 21: Popularity

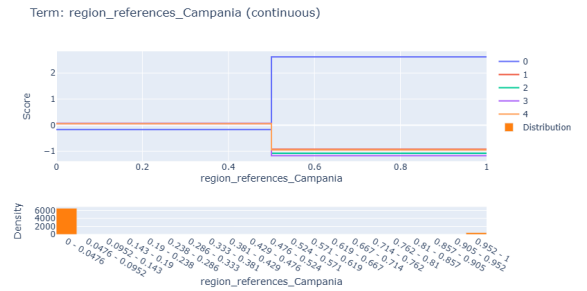


Figure 22: Campania References

In figure 21 and 22 we have the corresponding plots of **popularity** and **region_references_Campania**.

It is difficult to clearly inspect the relation of popularity with each class, but we can see that the region references feature is working as expected, because it helps in predicting the corresponding class.

3.6.1 Model Interactions

Unfortunately the `interpret` library doesn't support plots for interactions of a multiclass task, but we managed to retrieve some additional information from local plots of selected test instances.

The analysis suggests that the model relies heavily on the interaction between `popularity` and `relative_popularity`, which might also explain their high feature importance scores. We can see an example of this in figure 23 and 24. We think that this interaction serves as a measure of the overall popularity of the artist.

Another recurring interaction is `popularity` and `year`.

We believe the model has learned to associate specific time periods to the rise in popularity of certain rappers from specific schools. Figure 24 illustrates an example of the usage of this interaction.

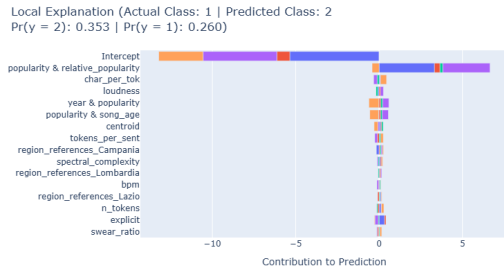


Figure 23: Misclassified Instance

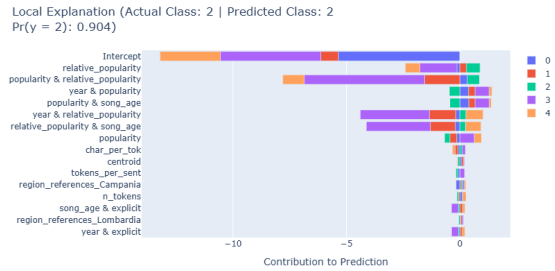


Figure 24: Correctly Classified Example

4 Task 4: Time Series Analysis

In this chapter, we address **Task 4**, analyzing the audio content of **Fedez** and **Fabri Fibra** to explore stylistic groupings and authorship. Raw audio presents unique challenges regarding the *semantic gap* and *extreme dimensionality* (approx. 8 million points per track). Crucially, unlike traditional time series, audio resists **aggressive resampling**: drastically reducing the sampling rate destroys high-frequency spectral information, eliminating the timbre and texture required for meaningful analysis.

4.1 Clustering & Feature Exploration

Given the computational intractability of raw audio, we shifted focus to compact latent representations, evaluating architectures designed to encode diverse signal dimensions. To identify the optimal basis for unsupervised grouping, we compared five feature sets: a low-level timbral baseline (**MFCC**), speech-optimized embeddings for phonetic flow (**Wav2Vec2**), music-specialized Transformers for harmonic structure (**MERT**), and

high-level semantic representations (**CLAP**). Additionally, a **Hybrid** model concatenating MERT and Wav2Vec2 was evaluated to synergize vocal delivery with instrumental complexity in a single holistic representation.

4.1.1 Post-Processing

To ensure robust clustering, embeddings were processed through a standardized chain to unify all tracks to 500 time steps. This involved: (i) **Gaussian Smoothing** ($\sigma = 2.0$) to mitigate aliasing—preventing high-frequency noise from being misinterpreted as structural patterns—and ensure the signal faithfully represents the latent trajectory; (ii) **Adaptive Average Pooling**, preferred over standard resampling to avoid artificial transitions or data loss. By aggregating mean activations within dynamic windows, it ensures that transient but semantically significant features are integrated rather than discarded; and (iii) **Magnitude Normalization** to eliminate magnitude bias, ensuring the algorithm prioritizes qualitative structural profiles over absolute intensity.

4.1.2 Clustering Strategies

We explored two distinct paradigms to group the tracks based on the aligned multi-modal embeddings, targeting both global characteristics and temporal shapes.

Static Approach (GAP): This method utilizes Global Average Pooling to collapse the temporal dimension into a single mean vector per song. These representations are reduced via PCA (retaining 90% of the variance) and grouped using K-Means. This strategy effectively clusters tracks based on their overall average "sonic signature".

Dynamic Approach (Temporal): To identify similarities in temporal evolution, we preserved the sequential nature of the data. Frames were reduced to 5 components (via PCA or UMAP) and normalized through Trajectory Scaling (`TimeSeriesScalerMeanVariance`). Clustering was performed using `TimeSeriesKMeans` with the Euclidean metric, which serves as an efficient proxy for shape similarity under rigorous normalization.

For both paradigms, the optimal number of clusters ($k = 4$) was identified and validated through the Elbow Method, ensuring a consistent basis for comparison between the static and dynamic results.

4.1.3 Evaluation Framework

To validate that clusters represent coherent musical styles, we implemented a hybrid protocol combining visualization, geometric metrics, and semantic assessment.

Visualization Strategy: We employed *UMAP* to qualitatively inspect the partitions. Static vectors were projected into 2D *Scatterplots* to assess separation, while dynamic features were reduced to a single dimension ($[T, F] \rightarrow [T, 1]$) to plot *Latent Trajectories*, visualizing shared structural patterns (e.g., drops, build-ups) within each group.

Semantic Validation (CLAP): To bridge the gap between numerical data and interpretation, we developed a zero-shot profiling pipeline using CLAP. By computing softmax probabilities between audio and a taxonomy of text prompts (e.g., *Mood*, *Genre*), we generated semantic profiles for each cluster. Intracluster coherence was quantified via a novel

Semantic Silhouette based on the **Jensen-Shannon Divergence** (JSD), ensuring consistent semantic distributions independent of signal magnitude.

Quantitative Metrics: Internal geometric quality was assessed using the **Silhouette Score**, confirming the $k = 4$ partition. External alignment with ground truth artist labels was measured through **ARI** and **NMI** to quantify the correlation between unsupervised stylistic clusters and artist identity.

4.1.4 Experimental Results: Global Average Pooling

Consistent with the Elbow Method ($k = 4$), *CLAP* emerged as optimal. Despite moderate geometric separation (Silhouette: 0.137), it achieved robust semantic coherence (0.105) and significant artist alignment (ARI: 0.17, NMI: 0.13). This is epitomized by **Cluster 2** ($> 90\%$ Fabri Fibra), characterized by "*Dry Voice*" and "*Fast Tempo*". These features accurately reflect the artist’s raw technical style in contrast to the heavily processed vocals of other groups, validating the unsupervised capture of sonic signatures.

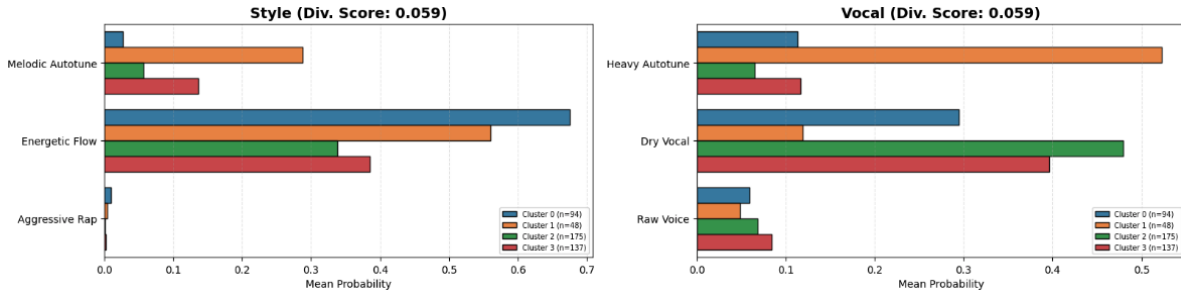


Figure 25: **Semantic Profiling.** Mean CLAP probabilities reveal distinct identities: Cluster 1 is defined by "*Melodic Autotune*" and Cluster 0 by "*Energetic Flow*", while the Fibra-exclusive Cluster 2 lacks these modern pop-trap features.

4.1.5 Experimental Results: Temporal Clustering

Temporal analysis yielded low geometric scores across all models (≈ 0.01), reflecting the complexity of aligning high-dimensional trajectories. However, a clear divergence in practical performance emerged.

CLAP : Despite its static success, CLAP failed in the dynamic domain. While cluster centroids exhibit distinct RMS energy profiles (Figure 26), this visual separation proved deceptive. The model could not group tracks coherently, yielding negligible external alignment (ARI: 0.02, NMI: 0.03) and indicating a lack of frame-level structural granularity.

The Hybrid Winner (MERT + Wav2Vec2): Conversely, the Hybrid architecture with UMAP emerged as the most effective strategy. Despite low internal compactness, it achieved superior external discrimination (ARI: 0.12, NMI: 0.16). As shown in Figure 27, it successfully creates clusters dominated by distinct artist signatures rather than random noise.

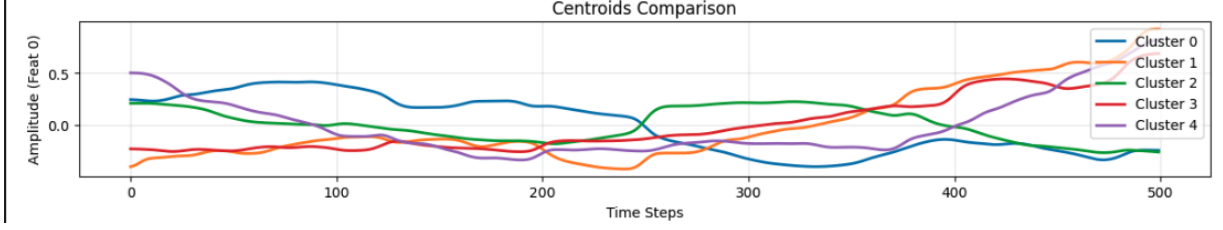


Figure 26: **CLAP Centroids**. Distinct RMS trajectories suggest dynamic variety, yet the model fails to group tracks coherently (ARI: 0.02), indicating a mismatch between visual centroid separation and meaningful clustering.

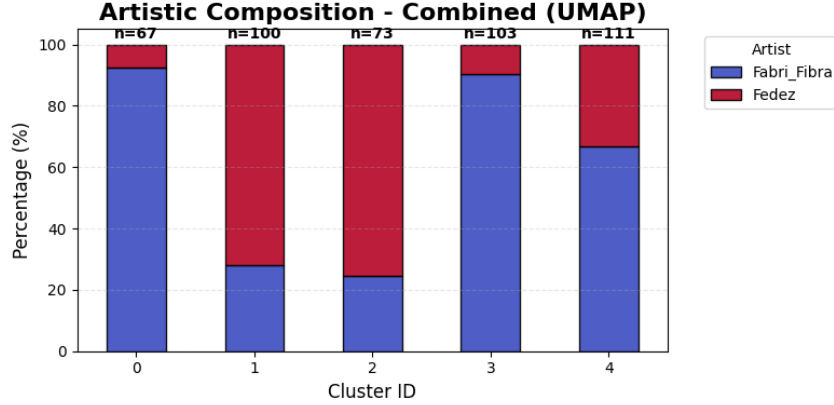


Figure 27: **Artist Composition (Hybrid Temporal)**. The Hybrid model (UMAP) achieves the best dynamic separation (**ARI: 0.12**), identifying clusters with strong artist-specific dominance (e.g., Cluster 0 is > 90% Fabri Fibra).

4.2 Cluster-wise Motif & Discord Discovery

Having identified the optimal partition using CLAP (GAP + PCA), we shifted focus to the internal granular structure of each discovered group. The goal was to extract *Motifs* (recurring patterns) and *Discords* (anomalies) that are not just local to a single track, but statistically representative of the specific cluster’s collective sonic identity. To achieve this, we applied time series data mining techniques based on the *Matrix Profile*.

Preprocessing: To isolate macroscopic structure, we extracted the *RMS energy envelope* of the tracks within each cluster. These were processed via the standardized pipeline defined in Sec. 4.1 (Gaussian Smoothing and Adaptive Average Pooling) but, crucially, using *Standard Scaling* to preserve relative dynamic excursions essential for intensity-based mining.

Methodology: To analyze the cluster as a unified entity, we concatenated the preprocessed envelopes of its member tracks into a single continuous sequence. Crucially, we inserted *NaN spacers* (logical separators) between segments. This strategy serves a dual purpose: concatenation allows the mining algorithm to scan the entire group globally to find shared signatures, while the spacers act as "walls" that prevent the detection of spurious matches across the artificial boundaries of different songs.

Motif Discovery: By computing the *Matrix Profile* on this concatenated super-sequence,

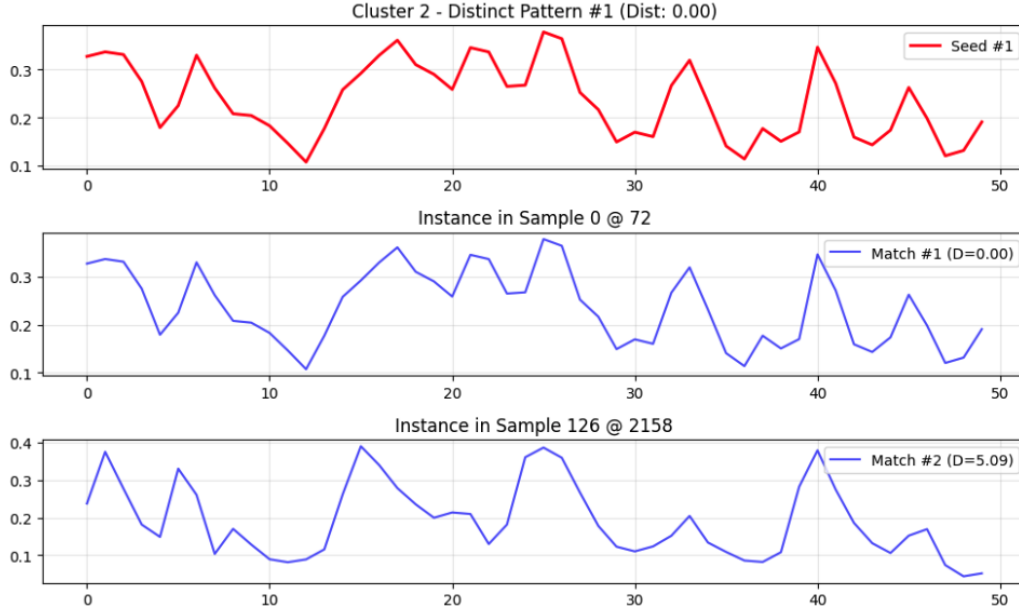


Figure 28: **Granular Analysis (Cluster 2)**. The top motif displays high-density, "nervous" oscillations. This jagged structure physically validates the *"Fast Tempo"* and *"Dry Voice"* tags, capturing the sharp syllabic articulation typical of technical rapping.

we identified the most significant repeating patterns (*Motifs*). To ensure these were true shared signatures rather than local repetitions (like a chorus repeated in the same song), we selected only those patterns appearing in at least two different tracks. This allowed us to pinpoint the specific musical segments that are most similar across the artist’s discography, effectively capturing the core "vibe" of the cluster.

Anomaly Detection (Discords): Complementing the search for repetitions, we analyzed the *maxima* of the Matrix Profile to identify *Discords*. For this task, we increased the window length to capture macroscopic non-matching segments. This allowed us to isolate structural outliers—such as unique bridges, spoken skits, or experimental sections—that possess no sonic parallel within the entire cluster.

4.3 Supervised Shapelet Analysis

To achieve granular explainability, we shifted focus from deep embeddings to five interpretable descriptors: *RMS*, *Spectral Centroid*, *Bandwidth*, *Rolloff*, and *ZCR*. These signals underwent the standardized smoothing and pooling pipeline (Sec. 4.3) to isolate macroscopic structure. We adopted **Balanced Accuracy** as the primary metric to rigorously assess generalization across artist classes, mitigating dataset imbalance. Two complementary strategies were employed:

4.3.1 Approach 1: Shapelet Transform & Classification

We utilized the *Random Shapelet Transform (RST)* to efficiently map raw time series into a discriminative "distance space" via stochastic sampling, avoiding the computational cost of brute-force search. These vectors fed a comparative benchmark (RF, SVC, KNN, XG-Boost), where the *Random Forest* emerged as the optimal architecture (**Balanced Acc:**

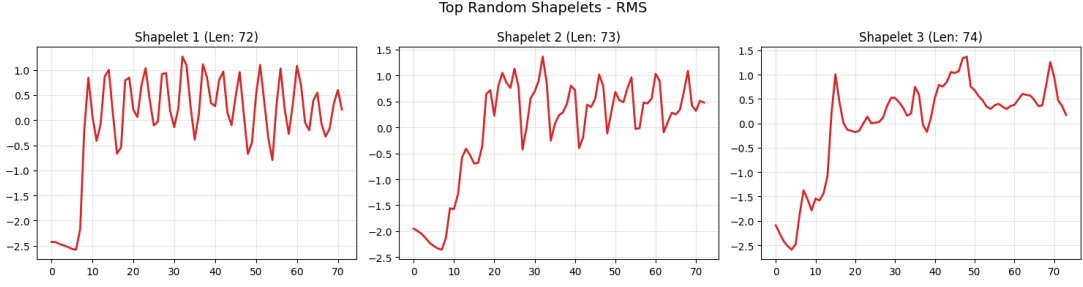


Figure 29: **Top Random Shapelets - RMS.** The figure illustrates the top three shapelets extracted by the Random Forest algorithm using the RMS feature. Each plot represents a short temporal energy pattern (varying in length between 72 and 74 samples) that the model identified as highly discriminative for classification.

0.71 in test, **Accuracy: 0.75**). Crucially, **RMS Energy** proved to be the superior discriminator across all models. This quantitative dominance suggests that the primary stylistic divergence lies in **dynamic flow** and rhythmic intensity—how the artists modulate energy over time—rather than in pure spectral timbre.

4.3.2 Approach 2: Multivariate Learning Shapelets

Complementing the univariate analysis, we adopted a **Multivariate Learning Shapelets** paradigm. Unlike Approach 1, which isolated descriptors, this strategy processes the full multi-channel signal simultaneously. By leveraging the differentiable *soft-minimum* distance, the model *synthesizes* optimal subsequences via gradient descent. This enables the discovery of complex, multi-dimensional signatures, capturing how dynamic intensity correlates with timbral evolution. Crucially, to prevent the optimization from collapsing into a trivial majority-class predictor—a risk present even with moderate dataset imbalance—we utilized a *WeightedRandomSampler* to enforce class equilibrium within training batches.

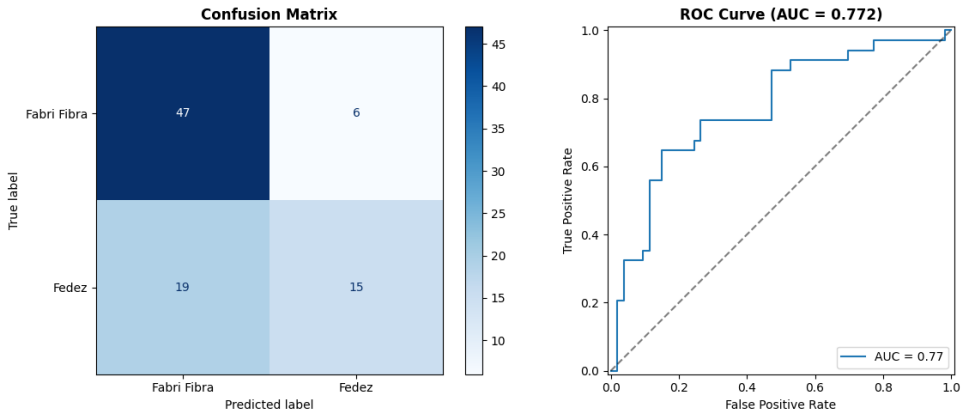


Figure 30: **Multivariate Learning Performance.** Despite a robust **ROC AUC of 0.77**, the gap between **Accuracy (0.71)** and **Balanced Accuracy (0.66)** highlights a residual bias towards the majority class (Fabri Fibra). Even with weighted sampling, the gradient-based synthesis slightly favored the dominant artist over strict class equilibrium.

5 Ethical and Legal Implications

5.1 Data Access and Copyright

Since the dataset has been provided to us by the professors, we assume that its creation adheres to its sources specific Terms of Services (ToS). Since the lyrics might be protected by copyright we are not allowed to redistribute the raw dataset, consequently the lyrics analysis is restricted to derivative insights.

5.2 Data Preprocessing: Scraping

During the data preparation phase, some missing metadata were retrieved through limited scraping activities. In particular, we queried public APIs and knowledge bases such as Spotify (via `spotipy`), MusicBrainz, and Wikidata to complete incomplete or missing information. This process was restricted exclusively to metadata and did not involve scraping or storing song lyrics.

These sources are explicitly designed to support programmatic access, which reduces the risk of violating terms of service when compared to uncontrolled web scraping. Moreover, the retrieved information was limited to fields strictly necessary for the task, in accordance with the principle of data minimization. Requests were performed in a controlled manner, respecting rate limits and avoiding excessive access. The purpose of this preprocessing step was solely to improve data quality and consistency, and not to perform individual-level profiling.

5.3 Personal Data and GDPR Compliance

Although song lyrics may reference religious or political beliefs, they are an artistic expression rather than an explicit declaration from the artist. As such, they do not constitute special category personal data under the GDPR. Metadata such as artists' birthplaces and dates of birth fall under the category of personal data, but not sensitive personal data. For these reasons we would apply techniques of pseudonymization rather than full anonymization.

For instance we could replace all artists' names with their IDs and use more general birthplaces and dates. Furthermore, it is important to use these data only for high-level statistical trends, avoiding individual-level profiling.

5.4 Clustering Analysis

Since the clustering has been performed on the tracks level, we are not dealing with personal data, so the risk of profiling individuals is very low.

It is worth noting that the feature **aggressiveness**, that has been used for the clusters interpretation, is just a mathematical proxy for specific musical and lyrical characteristics, it is not a legal or moral judgment on the tracks content or the artists.

5.5 Predictive Analysis: Regional Bias Risk

A critical ethical concern arises from the classification model’s reliance on popularity metrics. In the Italian music market, high popularity is often correlated with the industrial hubs of the Northern regions (e.g., Milan). If the model uses the interaction of popularity and relative popularity as a proxy for an artist overall popularity, thus biasing Northern regional origin, it risks reinforcing this bias.

Suggesting that "successful" rap is inherently a product of big cities or Northern regions could marginalize the work of artists from Southern or rural regions. The best course of action is to adjust the weights of these features and expand the dataset as much as possible, in order to make the musical and lyrical features more prominent. We don’t want to exclude the features related to popularity because their complete elimination hinders significantly the performance of the model. It’s important to find a balance between the inherent bias of the modern market and the absence of discrimination, which would also make the model more robust with respect to future trend shifts.

6 Ethical Implications of Time Series Analysis

In the context of our Time Series Analysis (Task 4), we processed *Audio* data, which is classified as a primary source of “digital traces” used to infer human behavior.

The exploration of latent representations using Deep Learning models (MERT, Wav2Vec2) introduces significant privacy risks, here are some of the major ones identified by us:

Biometric Data Classification: These models encode high-fidelity spectral information (timbre, pitch) which constitutes Biometric Data (GDPR Art. 9). As defined in the regulation, processing such data for unique identification falls under “Special Categories of Personal Data”, requiring stricter safeguards than standard metadata.

Black Box & Explainability: Neural networks act as “Black Boxes”, obscuring the decision-making process. This opacity hinders the Right of Explanation, making it impossible to determine if the model is classifying based on musical style or on the artist’s biometric identity.

Our Supervised Shapelet Analysis instead, implements *Privacy by Design* through active *Data Minimization*:

By extracting motifs solely from the RMS Energy envelope, we discard the spectral voiceprint. This ensures the model learns behavioral dynamics (rhythm/flow) instead of physiological traits. This abstraction removes biometric identifiers, shifting from high-risk biometric processing to privacy-preserving stylistic analysis.

Future Recommendation: To further align with the Differential Privacy standards presented in the course, future deployment should introduce noise injection (e.g., Laplace mechanism) in the shapelet distance calculation to provide mathematical guarantees against singling-out attacks.